Availability of renewable energy. To achieve the Paris Agreement targets of containing the rise in global temperature to 1.5°C to combat the climate change and mitigate the global warming, the demand for GH is expected to grow to 170 million tons by 2030 and 600 million tons by 2050. To support the demand, accelerating the deployment of the renewable energy sources is a big challenge.

Availability of water. Electrolysers using the electric energy produced by the renewable sources, such as sun and wind, split water into hydrogen and oxygen and this the most popular method of producing GH. However, electrolysis of water requires a large amount of water. Water scarcity in many regions poses a significant challenge.

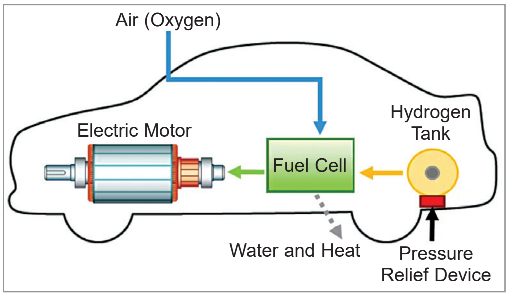
Research and development efforts. Widespread adoption of GH faces multiple technical issues ranging from improving the electrolysis technology; reducing the energy consumption by improving the overall efficiency of the electrolysers; developing more robust materials and designs to extend the lifespan of electrolysers; developing cost-effective and safe storage and transportation solutions; and reducing the initial capital costs and cost of renewable energy generation. To address these challenges a concerted R&D effort is needed from the governments, industry, research and development community, and other stakeholders.

Infrastructure development. The infrastructure for facilities, such as electrolysis, storage, transportation and refuelling are underdeveloped compared to other energy carriers, such as natural gas or electricity. The infrastructure required for integrating the GH into the existing energy systems also needs to be built. Development of the infrastructure requires significant investment by the governments, industry, and other stakeholders.

Regulatory framework. Developing and implementing a supportive regulatory framework for GH is much required. Governments need to establish and implement policies and regulations related to emission reduction targets, carbon pricing and trading, financial incentives for projects, and easy export of electrolysers and GH.

Box 1: Hydrogen fuel cells

Hydrogen fuel cells use the chemical energy of hydrogen to provide heat energy for buildings, fuel for transportation, and electricity for electrical vehicles and electronic devices. Fuel cells, which work like batteries but do not run down or need recharging, have two electrodes, a negative anode and a positive cathode, sandwiched around an electrolyte. A fuel, such as hydrogen, is fed to the anode, while air is fed to the cathode. The reaction to produce electricity takes place at the electrodes, the electrolyte carries the charged particles between the electrodes and a catalyst speeds up the reaction. The electrons go through the external circuit, creating a flow of electricity. The protons migrate through the electrolyte to the cathode, where they reunite with oxygen and the electrons to produce water and heat. The GH fuel cells are completely carbon-free with water as the only by-product.



Hydrogen fuel cell vehicle (Source: www.firehouse.com)

Opportunities

There are several promising GH opportunities available, and the future looks bright. GH is emerging as the most promising solution for transition to a cleaner, greener, and sustainable future in today's energy landscape. It is light, storable, energy-dense, produces no direct pollutants or greenhouse gases, and can be adopted in sectors difficult to decarbonise, such as transport, buildings, and power generation.

The pursuit of a sustainable and carbon-neutral future has never been more critical than it is today. The world faces the pressing challenges of environmental degradation, climate change, resource depletion, and the need to transition to clean and renewable energy sources. A huge transformative opportunity is emerging for adopting innovative solutions, developing infrastructure, framing supportive policies, and undertaking the research and development. To harness these opportunities, it is crucial to work collaboratively on a global scale to maximise the potential. Here is the key GH opportunity area.

Decarbonising the industrial sector. GH is a clean and efficient energy source and can play a crucial role in decarbonising the industrial sector which is difficult to decarbonise and is a major contributor to CO2 emissions. Industries such as steel, chemicals, and cement can substitute fossil fuels with GH to reduce their carbon footprints, meet sustainability targets, comply with the stringent regulations, and enhance their environmental performance.

Fuel for transportation. Hydrogen fuel cells (see Box 1) offer a practical and environment friendly alternative to internal combustion engines and electric vehicles with lithium-ion batteries. Use of GH as a clean and sustainable fuel in the transport sector can reduce the greenhouse gas emissions significantly. Moreover, GH can produce synthetic fuels, such as green ammonia and synthetic methane, which can be used in aviation and heavy-duty long-haul transportation.

Decarbonising the energy systems. It is possible to significantly reduce greenhouse gas emissions and accelerate the transition to cleaner energy systems by integrating GH into the energy mix. To address the energy intermittency, excess energy generated during the periods of high renewable energy